

Math 340: Elementary Matrix and Linear Algebra

PRACTICE MIDTERM ONE

Name: _____ Section#: _____ NetID: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of seven problems on five pages; the exam is 4 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process.
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (2+4+4 points) Let

$$A = \begin{bmatrix} 3 & -1 \\ 2 & 7 \\ 4 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 0 \\ 1 & 4 \\ 5 & -3 \end{bmatrix}, \quad \text{and } C = \begin{bmatrix} 4 & -6 \\ 0 & 2 \end{bmatrix}$$

be three matrices. Compute the following (if possible). If the computation is not possible, explain why not.

(a) $3B - A$

(b) $B^T A - C$

(c) $C^T B$

2. (4+3+4 points) Let

$$D = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 3 & 7 & 1 \end{bmatrix}, \quad \text{and } H = \begin{bmatrix} -4 & 0 & -4 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

be three matrices.

(a) Find matrices E and F such that $ED = G$ and $FD = H$.

(b) Find E^{-1} .

(c) Find $(EF)^{-1}$.

3. (3+3+4 points) Let

$$K = \begin{bmatrix} & 1 & 0 & 1 \\ a_1^2 - a_1 & 2 & a_2^2 \\ & 1 & 9 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 0 & b_1 & 2 \\ 1 & 0 & 3 \\ -2 & b_2 & 0 \end{bmatrix}, \quad \text{and } M = \begin{bmatrix} 2 & 1 \\ 0 & 0 \\ 6 & 3 \end{bmatrix}$$

be three matrices.

(a) Find all pairs (a_1, a_2) which make K symmetric.

(b) Find $b_1 + 4b_2$ if L is skew symmetric.

(c) Let f be a function defined by $f(\mathbf{x}) = M\mathbf{x}$. What is the domain, the codomain, and the range of f ?

4. (4+4+3 points) Let

$$N = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 3 & 4 & 0 \end{bmatrix} \text{ and } P = \begin{bmatrix} 1 & 4 & 3 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 2 & -8 \end{bmatrix}$$

be two matrices.

(a) Find N^{-1} , if it exists.

(b) Compute the reduced row echelon form of P .

(c) Identify all solutions to the linear system $P\mathbf{x} = \mathbf{0}$.

5. (2+6 points) Solve the following linear system

$$\begin{array}{rcrcrcrcrcrcl} 3x & - & 5y & + & 2z & = & 3 \\ -2x & + & 4y & + & 2z & = & -6 & . \\ 4x & - & 7y & + & z & = & 5 \end{array}$$

(a) Express this linear system in the form $A\mathbf{x} = \mathbf{b}$.

(b) If the system has a unique solution, provide it. If it has infinitely many solutions, provide an explicit parametrized form for these solutions. If it has no solutions, state this clearly. You must use Gaussian elimination or Gauss–Jordan reduction.

6. (3+3+4 points) Let

$$A = \begin{bmatrix} 5 & -a \\ a-2 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 8 & -9 & 1 & 0 \\ 6 & -7 & 3 & -3 \end{bmatrix}$$

be two matrices.

(a) Find a if $\det(A) = 15$.

(b) Compute $\det(B)$.

(c) The matrix C is obtained from B by the following row operations:

- (1) interchange rows 1 and 3,
 - (2) multiply row 4 by $\frac{1}{6}$,
 - (3) add 2 times row 3 to row 1,
 - (4) multiply row 1 by -2 ,
 - (5) subtract 5 times row 2 from row 4.
- What is $\det(C)$?

7. (4+4+5 points) Let

$$D = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -2 & 2 \\ 1 & 3 & -1 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 0 & 1 \\ 1 & b & 2 \\ 0 & 1 & 1 \end{bmatrix}, \quad \text{and } H = \begin{bmatrix} 2 & c & 5 \\ 0 & 1 & 4 \\ 8 & -2 & 0 \end{bmatrix}$$

be three matrices.

(a) Find b if $\det(DG) = 40$.

(b) Let $\mathbf{b} = \begin{bmatrix} 13 \\ 10 \\ 2 \end{bmatrix}$. In one of the solutions of the linear system $H\mathbf{x} = \mathbf{b}$, $x_1 = 0$, and in another solution, $x_1 = -3$. Find c .